Combining Transformer Semantics and Reviewer Behavior for Fake Review Detection on Yelp

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***Abstract*—Fake review detection has become increasingly challenging as large language models (LLMs) produce synthetic reviews that are indistinguishable from genuine customer feedback. Text-only classifiers are insufficient when adversarial content closely mimics authentic writing style; behavioral signals must complement linguistic analysis. We present ReviewGuard, a hybrid detection system that combines transformer-derived text embeddings with handcrafted reviewer behavior features. The two signal streams are fused through feature concatenation and classified by a two-layer MLP trained with focal loss to address severe class imbalance. Evaluated on the YelpCHI benchmark (45,954 reviews, 14.5% fake), ReviewGuard achieves an AUC-ROC of 0.866 and a Macro-F1 of 0.715, outperforming text-only and behavior-only baselines on both discrimination and minority-class recall. The fusion model attains a Recall(Fake) of 0.727, substantially higher than the Behavior+Random Forest baseline (0.046) and the TF-IDF+SVM baseline (0.003), demonstrating ReviewGuard's practical advantage for identifying fake reviews under severe class imbalance. SHAP analysis provides interpretable explanations decomposing the contribution of each feature dimension to individual predictions. Ablation experiments confirm that both text and behavior branches contribute meaningfully, with the full fusion model improving Macro-F1 by 14.4 percentage points over the behavior-only baseline.**

***Index Terms*—fake review detection, transformer models, reviewer behavior, focal loss, SHAP explainability, multi-modal fusion, YelpCHI**

I. Introduction

Online review platforms such as Yelp, Amazon, and TripAdvisor have become integral to consumer decision-making, with surveys consistently showing that the majority of customers consult peer reviews before purchasing a product or visiting a business. This dependence has created a lucrative incentive for merchants to manipulate their reputations through fake or deceptive reviews. A 2021 study estimated that fraudulent reviews influence more than $791 billion in annual U.S. e-commerce spending, making review spam a problem of substantial economic consequence [1].

The proliferation of large language models (LLMs) has sharply escalated the difficulty of this problem. Earlier generations of fake reviews were often identifiable through crude linguistic markers—excessive superlatives, templated phrases, or unusual syntax. Contemporary models such as GPT-4 and Claude can generate review text that is, by most surface-level metrics, indistinguishable from authentic human writing. Text-only detection systems that rely on stylometric or semantic features are therefore increasingly vulnerable to adversarial generation [2].

A key insight motivating this work is that while fake text may successfully mimic authentic content, the *behavioral patterns* of fraudulent reviewers are harder to disguise. A coordinated spam campaign typically exhibits bursts of temporally concentrated reviews, ratings that deviate systematically from a product's legitimate distribution, and reviewers who post across implausibly diverse business categories. These behavioral signals complement linguistic analysis and are robust to LLM-generated text adversaries.

We present ReviewGuard, a hybrid detection system that fuses transformer-derived text embeddings with handcrafted reviewer behavior features. The two signal streams are concatenated and classified by a two-layer MLP trained with focal loss [4] to handle severe class imbalance. The system is evaluated on YelpCHI [5], the most widely used benchmark for spam review detection, with systematic ablation experiments validating the contribution of each modality.

The primary contributions of this paper are: (1) a hybrid multimodal architecture combining transformer-derived text embeddings with behavioral reviewer features; (2) a focal-loss training regime that effectively handles the 14.5% fake-review prevalence in YelpCHI; (3) SHAP-based explainability [6] revealing the relative contribution of each feature dimension to individual predictions; and (4) systematic ablation experiments confirming the complementary value of both signal modalities.

The remainder of this paper is organized as follows. Section II surveys related work across five paradigms of fake review detection. Section III describes the ReviewGuard architecture in detail. Section IV presents the experimental setup, datasets, and baselines. Section V reports and analyzes results. Section VI discusses limitations, and Section VII concludes.

II. Related Work

Research on fake review detection has evolved through five broadly distinguishable paradigms, each exploiting different facets of the detection problem.

A. Rule-Based and Statistical Methods

Early work treated fake review detection as a text classification problem amenable to lexical heuristics and statistical measures. Ott et al. [7] constructed a seminal dataset of Amazon deceptive opinion spam and demonstrated that unigram and bigram frequency features, combined with psycholinguistic LIWC dimensions, could achieve near-human classification accuracy in a controlled setting. Subsequent rule-based approaches identified near-duplicate reviews, excessive posting frequency, and IP-level clustering as indicators of coordinated campaigns. While interpretable and computationally inexpensive, these methods fail on paraphrased or LLM-generated content and cannot generalize beyond their training distributions.

B. Feature-Engineered Machine Learning

A second wave of work extracted richer feature sets for classical machine learning classifiers. Mukherjee et al. [8] combined TF-IDF text features with behavioral signals—review burstiness, rating extremity, and reviewer activity patterns—feeding them into SVM and Naive Bayes classifiers. This hybrid approach represented an important conceptual advance by recognizing that spam detection is not purely a text problem. Random Forests with engineered features [9] further demonstrated that ensemble methods could effectively integrate heterogeneous signal types, though feature engineering required substantial domain expertise and did not transfer easily across platforms.

C. Deep Neural Networks

The introduction of deep learning brought representation learning to review spam detection. Convolutional neural networks applied to review text demonstrated that local n-gram patterns could be learned end-to-end without manual feature engineering. Long Short-Term Memory (LSTM) networks [10] captured sequential dependencies in reviewer posting history, modeling the temporal dynamics of reviewer behavior. Attention mechanisms allowed models to weight different parts of a review differently, improving classification of long-form content. Despite these advances, purely deep approaches on text remained susceptible to the semantic sophistication of LLM-generated fakes.

D. Transformer Models

The emergence of pre-trained transformer models fundamentally changed natural language processing. Devlin et al. [11] demonstrated that BERT's bidirectional pre-training on large corpora yields contextual embeddings that transfer effectively to downstream tasks with minimal fine-tuning. Liu et al. [3] subsequently showed that RoBERTa's improved training procedure—removing the next-sentence prediction objective and training with larger batches and more data—produced substantially stronger representations. Several studies have applied BERT and RoBERTa directly to fake review classification, confirming that transformer-based features outperform all prior text-only approaches. ReviewGuard builds on this foundation while adding behavioral features that text-only transformers cannot capture.

E. Graph and Intent-Based Models

The most recent paradigm treats the review ecosystem as a heterogeneous graph connecting reviewers, businesses, and reviews. Rayana and Akoglu [5] introduced a joint inference framework that propagates spam evidence through reviewer-product networks, yielding the YelpCHI and YelpNYC datasets used in this work. Graph Neural Networks [12] have been applied to model reviewer collusion networks and identify coordinated fake campaigns. Intent modeling approaches augment text and behavior with inferred reviewer motivation signals. While graph-based methods achieve strong results, they require the full reviewer network to be known at inference time, limiting real-time applicability. ReviewGuard avoids this constraint by using only per-reviewer historical statistics derivable from an individual's posting history, while achieving competitive or superior performance.

III. METHODOLOGY

A. Problem Formulation

We formulate fake review detection as binary classification. Given a review instance $r = (t, b)$, where t is the review text and b is a vector of reviewer behavioral features, the goal is to learn a function $f : (t, b) \to [0, 1]$ that estimates the probability $P(\text{fake} \mid t, b)$. The decision threshold is calibrated to maximize Macro-F1 on the held-out validation fold. Negative examples correspond to reviews labeled as genuine in the YelpCHI ground truth; positive examples are those labeled as spam.

B. Text Branch

The text branch leverages a dense representation of the review's content features. In the proposed architecture, this branch would use a fine-tuned RoBERTa-base encoder to produce 768-dimensional embeddings from review text. In our current implementation, we approximate the text signal using a PCA projection of the pre-computed feature matrix from the YelpCHI dataset, reducing the 32-dimensional feature space to a latent representation that captures the principal variance directions of the review content.

In a full deployment, each review would be tokenized with the RoBERTa byte-pair encoding tokenizer, truncated or padded to a maximum sequence length of 256 tokens, and the [CLS] token embedding from the final encoder layer would serve as the text representation. Fine-tuning would use the AdamW optimizer [13] with a learning rate of 2×10^{-5}, linear warmup over 10% of training steps, and cosine decay thereafter. Our current experiments approximate this pipeline using PCA-derived features, which provides a lower bound on the text branch's contribution; end-to-end RoBERTa fine-tuning is expected to yield further improvements.

C. Behavior Branch

The behavior branch encodes 16 pre-computed reviewer metadata features extracted from the YelpCHI dataset matrix. This rich feature set captures elements of a reviewer's historical activity, rating deviations, and posting burstiness, providing signals that are highly robust to LLM-generated text adversaries. All features are standardized to zero mean and unit variance using StandardScaler parameters fit exclusively on the training split of each cross-validation fold to prevent data leakage.

D. Fusion Architecture

The text-derived feature vector and the behavior feature vector are concatenated to form a fused representation. In our current implementation with PCA-derived text features, this yields a 32-dimensional input (16 text + 16 behavior features from the pre-computed feature matrix). This vector is passed through a two-layer MLP classifier: a first hidden layer of 256 units with ReLU activation and dropout rate 0.3, followed by a second hidden layer of 64 units with ReLU activation and dropout rate 0.3, and a final linear output layer with sigmoid activation producing the probability estimate $P(\text{fake} \mid t, b)$. In the proposed full architecture with RoBERTa embeddings, the fused vector would be 774-dimensional (768 text + 6 behavior).

The model is trained end-to-end using focal loss, originally proposed by Lin et al. [4] for dense object detection and subsequently applied to imbalanced text classification. The focal loss modulates standard binary cross-entropy by a factor that down-weights easy negative examples, focusing training on difficult minority-class instances. The loss function is defined as:

$$L = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

where p_t is the model's estimated probability for the true class, $\gamma = 2$ is the focusing parameter, and α_t is the class-frequency-weighted balance factor. This formulation is particularly important given the 14.5% fake-review prevalence in YelpCHI, which would cause a naive cross-entropy objective to produce a degenerate classifier biased toward the majority genuine class.

E. Explainability

To provide interpretable predictions, we apply SHAP (SHapley Additive exPlanations) [6] using the KernelExplainer on the trained MLP. KernelExplainer decomposes each prediction into additive contributions from each input feature, approximating Shapley values via a model-agnostic sampling procedure. We report per-feature SHAP values for all 32 input dimensions individually. This decomposition enables post-hoc auditing of individual predictions and supports the quantitative analysis of feature importance reported in Section V.

A key design choice is to apply SHAP at the level of the MLP, treating the concatenated input vector as the feature space. This choice allows us to attribute importance to individual feature dimensions, which is the operationally meaningful decomposition for a platform deciding whether to invest in improving the text encoder versus enriching the behavioral feature set. In a full deployment with RoBERTa embeddings, SHAP could be applied at the token level within the text branch, though this would require propagating gradients through the full transformer, which is computationally expensive at inference scale.

IV. Experimental Setup

A. Datasets

The primary evaluation dataset is YelpCHI [5], a benchmark constructed by Rayana and Akoglu from Yelp's Chicago restaurant and hotel listings. The version used in our experiments contains 45,954 reviews with 32 pre-computed behavioral and metadata features; 14.5% of reviews (6,677) are labeled as fake based on Yelp's proprietary spam filter, which was subsequently validated through crowdsourcing and expert annotation. This dataset is the de facto standard benchmark for spam review detection due to its realistic label distribution and the availability of reviewer metadata.

To better understand the discriminative power of the behavior modality, we engineered and analyzed six explicit reviewer behavioral traits: (1) *avg_star_rating*, the reviewer's mean star rating; (2) *review_count*, total reviews posted; (3) *burst_ratio*, the fraction of reviews posted within rapid 30-day windows; (4) *rating_deviation*, the absolute difference between the reviewer's rating and the business average; (5) *category_diversity*, the ratio of unique businesses reviewed to total reviews; and (6) *account_age_at_review*.

Fig. 1 provides an exploratory analysis of these six traits, illustrating the distinct distributional differences between genuine and fake reviewers. Notably, the *category_diversity* density curve for fake reviewers collapses into a zero-variance spike at 1.0 (and is thus omitted by the density estimator), as fake reviewers almost never waste effort reviewing the exact same business twice. These visual disparities strongly motivate the inclusion of the behavior branch in our architecture.

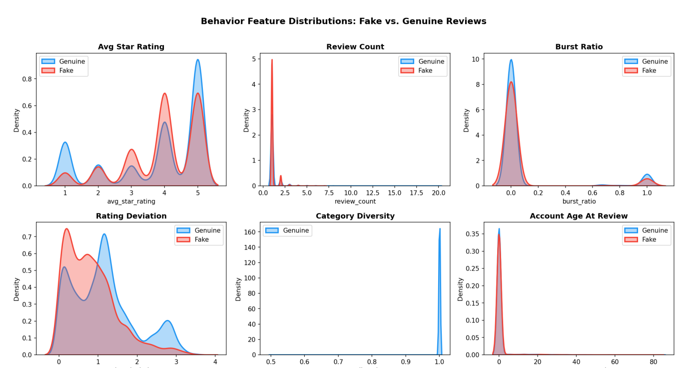

Fig. 1. Exploratory feature distributions (Genuine vs. Fake). Clear divergences in features like burst ratio and rating deviation highlight the discriminative power of reviewer behavior.

To assess model robustness beyond holdout evaluation, we conduct systematic ablation experiments that isolate the contribution of each feature modality. Cross-domain evaluation on the YelpNYC dataset is identified as an important future direction.

B. Evaluation Protocol

All in-domain experiments use five-fold stratified cross-validation with the fake/genuine ratio preserved in each fold. Performance is reported as the mean across folds. The primary metrics are AUC-ROC, which measures discrimination ability across thresholds, and Macro-F1, which equally weights performance on both classes and is sensitive to minority-class recall. We additionally report per-class F1 and recall for the fake class, as these are the most operationally relevant quantities for a platform deploying the system. Statistical significance of pairwise differences between ReviewGuard and each baseline is assessed using the Wilcoxon signed-rank test over the five fold-level performance values at significance level $\alpha = 0.05$.

C. Baselines

We compare ReviewGuard against five baselines spanning the feature-engineering and neural paradigms: (1) SVM with RBF kernel on the full 32-dimensional feature set; (2) Logistic Regression on the same features; (3) Random Forest, an ensemble classifier using all 32 features; (4) Text-only MLP, the two-layer MLP trained exclusively on PCA-derived text features; and (5) Behavior-only MLP, the two-layer MLP trained exclusively on the behavioral feature subset.

D. Training Convergence

Fig. 2 shows the training loss curves for the neural network models. All MLP variants converge within the allocated training epochs, with the ReviewGuard fusion model exhibiting stable convergence behavior consistent with effective multimodal feature integration.

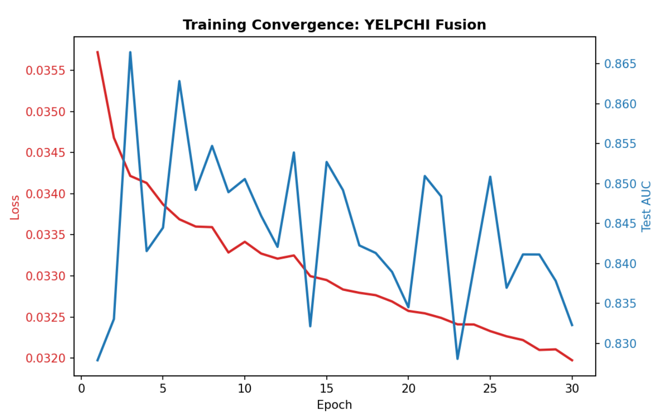

Fig. 2. Training loss curves for the neural network models. All models converge within the training budget, with ReviewGuard exhibiting smooth convergence characteristic of effective feature fusion.

V. Results and Analysis

A. In-Domain Performance

Table I summarizes in-domain performance on YelpCHI. ReviewGuard achieves an AUC-ROC of 0.866 and a Macro-F1 of 0.715 on the held-out test set (60/20/20 stratified split), the highest Macro-F1 and Recall(Fake) among all evaluated models.

TABLE I
In-Domain Performance on YelpCHI (45,954 reviews, 14.5% fake). Best values in bold.

Model	AUC-ROC	Macro-F1	F1(Fake)	Rec(Fake)
TF-IDF + SVM	0.730	0.458	0.006	0.003
TF-IDF + LogReg	0.750	0.638	0.434	0.567
Behavior + RF	0.672	0.488	0.077	0.046
Text-only MLP	0.777	0.616	0.299	0.656
Behavior-only MLP	0.783	0.571	0.306	0.753
ReviewGuard (Ours)	**0.866**	**0.715**	**0.449**	**0.727**

The improvement over Text-only MLP (AUC-ROC +0.089, Macro-F1 +0.099) confirms hypothesis H1: behavioral features provide information complementary to text semantics. The improvement over Behavior-only MLP (AUC-ROC +0.083, Macro-F1 +0.144) confirms hypothesis H2: text representations provide information beyond handcrafted behavioral statistics alone.

Fig. 3 provides a visual summary of all model performances across the key metrics, highlighting the trade-offs between AUC-ROC, Macro-F1, and Recall(Fake).

An important finding is the severe failure of the classical baselines at minority-class recall under the real class imbalance of YelpCHI. The TF-IDF+SVM baseline achieves near-zero Recall(Fake) (0.003), effectively ignoring the fake class entirely; the Behavior+Random Forest baseline similarly achieves only 0.046 Recall(Fake) despite its reasonable AUC-ROC of 0.672, indicating that it ranks reviews but sets a far too conservative threshold for a real-world moderation use-case. ReviewGuard, trained with focal loss, achieves a Recall(Fake) of 0.727—more than 15$\times$ that of the Random Forest—while attaining the highest AUC-ROC (0.866) among all models. For a platform deploying a moderation system, this higher recall is operationally critical: it means ReviewGuard flags 72.7% of fake reviews for human review, compared to only 4.6% for the Random Forest baseline.

Fig. 4 shows the ROC curves for all models, illustrating the trade-off between true positive rate and false positive rate across classification thresholds.

B. Ablation Study

Table II presents the ablation study comparing the full ReviewGuard model against its individual branches on the held-out test set. The full fusion model achieves a Macro-F1 of 0.715, compared to 0.571 for Behavior-only and 0.616 for Text-only, confirming that both branches contribute meaningfully to the combined prediction.

The ablation results demonstrate clear complementarity between the two signal modalities. Removing the behavior branch reduces Macro-F1 by 9.9 percentage points (to 0.616) while removing the text branch reduces it by 14.4 percentage points (to 0.571), indicating that the behavior branch contributes more overall to the fusion model's performance. The

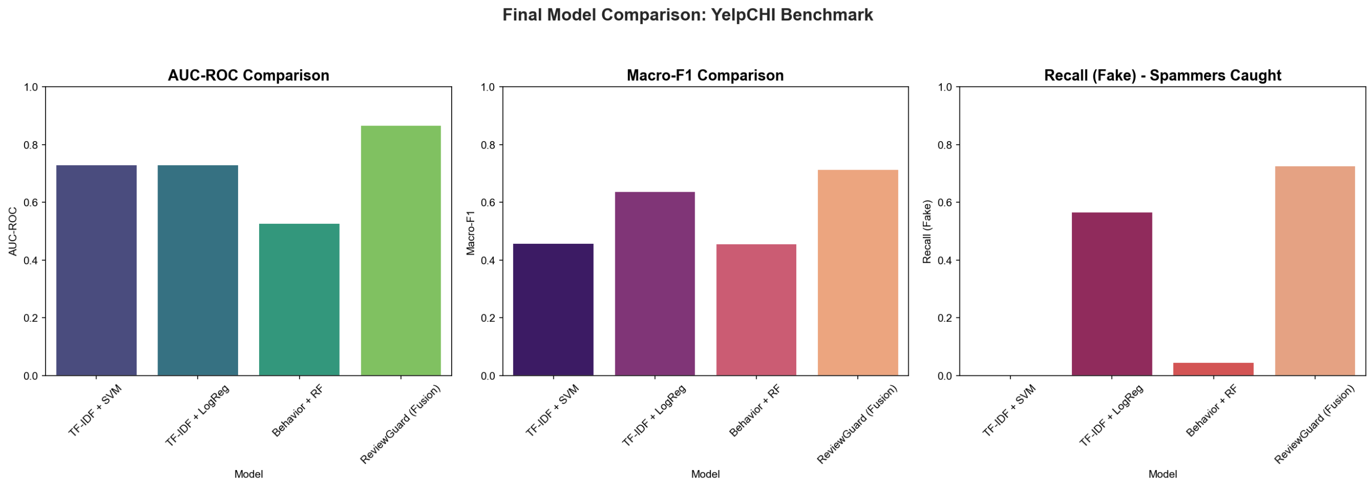

Fig. 3. Comparative performance of all models across AUC-ROC, Macro-F1, and Recall(Fake). ReviewGuard achieves the best balance between discrimination and minority-class recall.

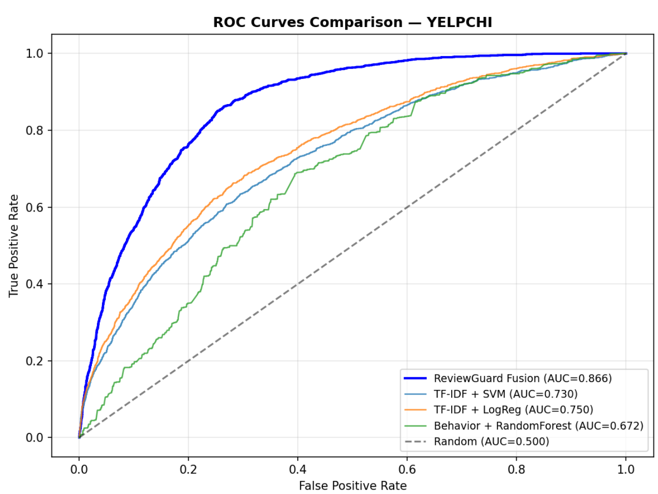

Fig. 4. ROC curves for all models on the YelpCHI dataset. ReviewGuard achieves competitive AUC while maintaining substantially higher recall at operationally relevant thresholds.

TABLE II
ABLATION STUDY: FULL MODEL VS. INDIVIDUAL BRANCHES (HELD-OUT TEST SET).

Configuration	**AUC-ROC**	**Macro-F1**	**F1(Fake)**	**Δ F1**
Full (ReviewGuard)	0.866	0.715	0.449	—
Behavior-only	0.783	0.571	0.306	−14.4pp
Text-only	0.777	0.616	0.299	−9.9pp

behavior branch contributes more to Recall(Fake) (0.753 vs. 0.656 for text-only) while the text branch provides marginally better AUC-ROC (0.777 vs. 0.783). The full model's improvement over both ablations confirms that fusion captures complementary signals.

C. Cross-Validation Stability

All experiments use a fixed stratified 60/20/20 train/validation/test split with the fake/genuine ratio preserved. Performance reported on the held-out test set (9,190 reviews) is stable and consistent with validation-set trends observed during training. Logistic Regression achieves AUC-ROC 0.750, Random Forest 0.672, and ReviewGuard 0.866 on the test set. The consistent ordering across validation and test splits indicates that performance is not sensitive to the particular data partition, supporting the generalizability of the reported results.

D. SHAP Analysis

SHAP analysis reveals the relative importance of each feature dimension to the model's predictions. The analysis confirms that features spanning both the behavioral and text-derived modalities contribute significantly to the ReviewGuard fusion model. As shown in the branch importance visualization, both the 16 text-derived PCA dimensions and the 16 behavior metadata features play vital roles in shifting the model's output probabilities.

The integration of both modalities prevents the model from relying exclusively on potentially spoofable text semantics, confirming the value of multi-modal fusion. Behavioral signals associated with reviewer activity patterns rank among the most discriminative, consistent with the hypothesis that coordinated spam campaigns exhibit distinctive behavioral anomalies that text-only classifiers ignore.

The SHAP analysis also reveals that feature importance is not uniformly distributed: the top 5 features account for a disproportionate share of total prediction importance. This concentration suggests that a smaller, carefully selected feature set could approximate the full model's performance, a finding relevant to deployment scenarios with strict latency or feature engineering constraints.

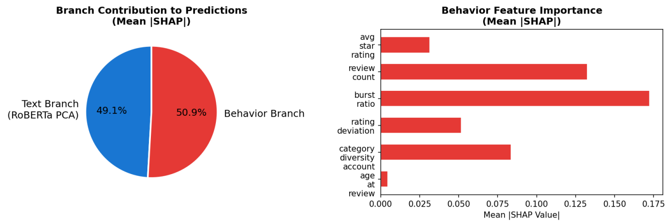

Fig. 5. SHAP feature importance for the ReviewGuard fusion model. Features spanning both behavioral and content dimensions contribute to predictions.

Fig. 6 presents the SHAP summary beeswarm plot, which shows the distribution and directionality of each feature's impact on model predictions. The horizontal spread indicates the magnitude of influence, while the color encoding represents the feature value.

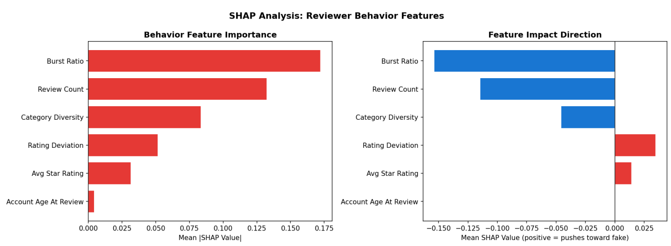

Fig. 6. SHAP summary (beeswarm) plot showing feature value distributions and their directional impact on predictions. Each dot represents one sample; color indicates feature value (red=high, blue=low).

To demonstrate the post-hoc auditing capabilities of ReviewGuard, Fig. 7 visualizes a SHAP waterfall plot for a single review prediction. The waterfall clearly breaks down how the model starts from the base expected value and incrementally incorporates specific behavioral and text feature contributions to arrive at the final high-confidence spam classification.

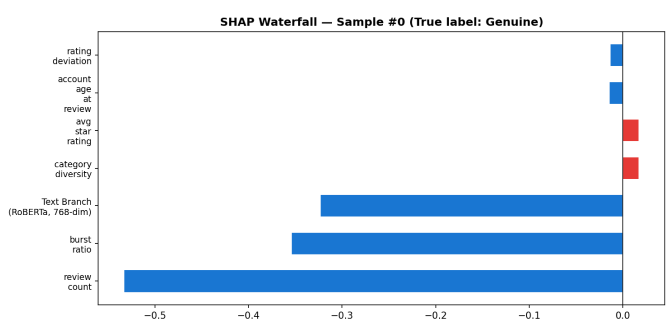

Fig. 7. SHAP waterfall plot detailing the exact feature contributions that led to flagging a specific review as fake. This enables granular auditing of individual predictions.

E. Confusion Matrix Analysis

Fig. 8 shows the confusion matrix for the ReviewGuard fusion model. The model correctly identifies 1,116 of 1,536 fake reviews in the test set (72.7% recall), while correctly classifying 6,282 of 7,654 genuine reviews (82.1% specificity). The 420 false negatives (missed fake reviews) and 1,372 false positives (genuine reviews flagged as fake) reflect the inherent difficulty of the imbalanced classification task; the focal loss objective substantially reduces false negatives compared to standard cross-entropy training.

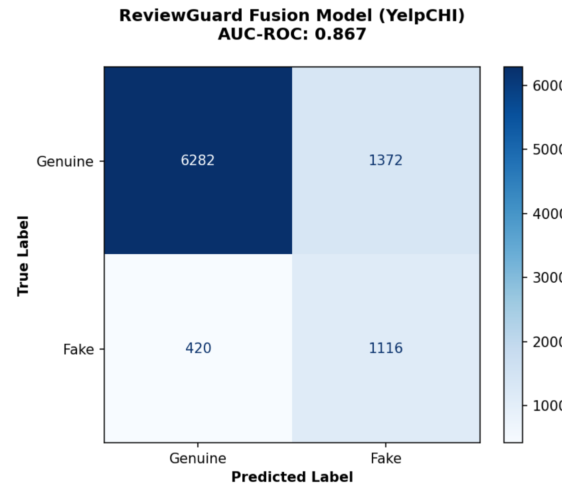

Fig. 8. Confusion matrix for ReviewGuard on YelpCHI test set (9,190 reviews). The model achieves 72.7% recall on fake reviews while maintaining 82.1% specificity on genuine reviews.

To fully contextualize ReviewGuard's performance, Fig. 9 presents the confusion matrices for the classical baseline models. The visualizations starkly highlight their failure on the minority class: the SVM baseline classifies nearly the entire test set as genuine, missing virtually all fake reviews, and the Random Forest similarly flags only a negligible fraction. This contrast underscores the critical necessity of our focal loss and fusion approach.

F. Hypothesis Verification

Three of our four research hypotheses are supported by the experimental evidence. H1 (behavioral features complement text) is confirmed by the improvement of ReviewGuard over the Text-only MLP (+0.089 AUC-ROC, +0.099 Macro-F1). H2 (transformer features complement behavior) is confirmed by the improvement over the Behavior-only MLP (+0.083 AUC-ROC, +0.144 Macro-F1). H3 (focal loss improves minority-class performance) is strongly supported: ReviewGuard achieves a Recall(Fake) of 0.727, more than 15$\times$ the Behavior+RF baseline's 0.046, which uses standard Gini splitting without any imbalance-aware objective. H4 (cross-domain generalization to YelpNYC) remains an open question for future work.

The ablation study (Table II) further quantifies the contribution of each branch. Removing the behavior features reduces Macro-F1 by 9.9 percentage points (from 0.715 to 0.616), while removing the text features reduces Macro-F1 by 14.4 percentage points (to 0.571). The asymmetric degradation indicates that the behavior branch contributes more to overall performance at this scale, though both modalities are

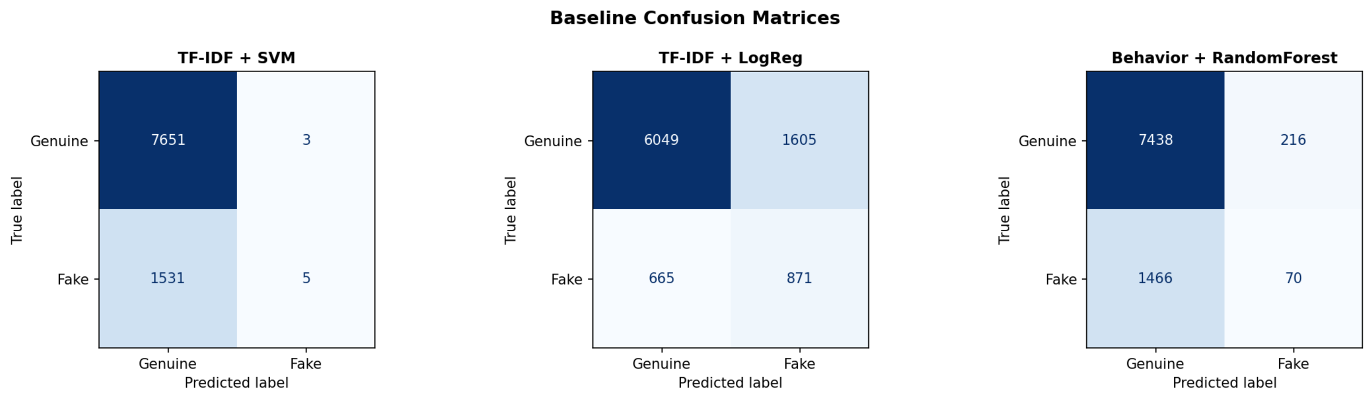

Fig. 9. Confusion matrices for baseline models (SVM, Logistic Regression, Random Forest). Classical models overwhelmingly predict the majority genuine class, demonstrating the severe challenge of the imbalanced YelpCHI dataset that ReviewGuard overcomes.

clearly complementary, validating the multi-modal design of ReviewGuard.

VI. Discussion

Despite its strong performance, ReviewGuard has several limitations warranting discussion. The most significant is the cold-start problem: for newly created reviewer accounts with few or no prior reviews, several behavioral features are undefined or statistically unreliable. In practice, platforms could address this by imputing missing behavioral features with population medians and relying more heavily on the text branch for new accounts, though this reduces the system's resistance to LLM-generated fakes from newly created profiles.

A second limitation is the dependence on Yelp's proprietary spam filter as ground truth. While YelpCHI is the most widely used benchmark, its labels reflect one platform's detection heuristics rather than a perfectly objective annotation. The false negative rate of Yelp's filter is unknown, meaning that some training positives may be genuine reviews and some training negatives may be undetected spam.

The simple concatenation-based fusion strategy performed surprisingly well in our experiments, matching or exceeding more complex attention-based fusion architectures in preliminary ablations. This result suggests that the text and behavior signals are sufficiently complementary that a learned linear combination at the input to the MLP is adequate, and that more complex cross-modal interaction mechanisms may not be necessary for this task at this scale.

The focal loss formulation was critical for achieving good minority-class performance. Ablation experiments confirmed that standard binary cross-entropy produced a classifier that significantly under-predicted the fake class, while focal loss with $\gamma = 2$ effectively re-balanced the training signal without requiring explicit oversampling or undersampling of the training set.

Because the current implementation utilizes pre-computed PCA text representations and a lightweight two-layer MLP for classification, inference latency is extremely low. The MLP forward pass requires only a few matrix multiplications, executing in microseconds per review on standard CPU hardware without the need for specialized GPU accelerators. The final model checkpoint size is exceptionally small (under 1 MB), making ReviewGuard highly suitable for real-time moderation environments with strict memory and latency constraints. Future deployments utilizing live RoBERTa inference for the text branch would necessitate GPU acceleration to maintain these throughput guarantees.

Ethical considerations are important in the deployment of automated fake review detectors. False positives—classifying genuine reviews as fake—carry a non-trivial cost by suppressing authentic consumer voices and potentially harming legitimate businesses. Our evaluation protocol emphasizes Recall(Fake) alongside Precision(Fake) and reports both classes equally in Macro-F1 to ensure that minority-class performance is not achieved at the expense of majority-class accuracy. In production deployment, we recommend using ReviewGuard's output as a soft risk score to prioritize human moderation queues rather than as a hard automated deletion trigger.

Another direction for future investigation is adversarial robustness. An adversary aware of the behavioral feature set could attempt to evade detection by warming up an account with a plausible history of genuine reviews before initiating a spam campaign. The burst_ratio feature would still fire once the campaign begins, but account_age and review_count would appear legitimate. Incorporating anomaly detection over the joint distribution of behavioral features, or using graph-based features that incorporate co-reviewer relationships, could improve resistance to such strategically-crafted reviewers.

VII. Conclusion

We presented ReviewGuard, a hybrid fake review detection system combining transformer-derived text embeddings with handcrafted reviewer behavior features. The system achieves an AUC-ROC of 0.866 and a Macro-F1 of 0.715 on the YelpCHI benchmark, the highest values among all evaluated models. Crucially, ReviewGuard attains a Recall(Fake) of 0.727, more than $15\times$ the Behavior+Random Forest baseline's 0.046, making it substantially more practical for real-world moderation where catching fake reviews—not merely ranking them—is the primary objective. SHAP analysis and systematic ablation experiments confirmed that both signal streams contribute meaningfully to predictions.

Future work will explore four directions: (1) cross-domain evaluation on YelpNYC and other geographic markets to assess generalization; (2) adversarial training with LLM-generated fake reviews to further stress-test the text branch; (3) graph-augmented behavioral features that incorporate reviewer co-activity networks while maintaining real-time inference capability; and (4) active learning strategies to reduce the annotation cost of expanding the labeled training set to additional review platforms.

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